



Project NLP

Automated Customer Reviews

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Introduction

This project focuses on classifying reviews and generating interactive blog posts based on real product reviews on the Amazon platform. The goal of the project is to build a machine learning model capable of classifying sentiments (positive, neutral, negative) for each review, as well as clustering reviews into categories and generating blog post texts based on the analysis.





01

Dataset&Data preprocessing



Dataset

A dataset containing customer reviews for Amazon products from Datafiniti was used. Specifically, a CSV file containing the following information:

- **Ratings (reviews.rating):** The rating given by the customer for the product.
- **Text (reviews.text):** The textual content of the review.
- **Name (name):** The name of products
- **Primary categories (primaryCategories):** The main category the product belongs to (e.g., Tablets, Accessories, Smart Home Devices).

Data preprocessing

● 1- Data Cleaning

The data was cleaned through:

- Dropping any missing values using **dropna()**.
- Simplifying the text by converting it to lowercase, removing non-text symbols, and correcting formatting using regular expressions.

● 2- Sentiment Classification

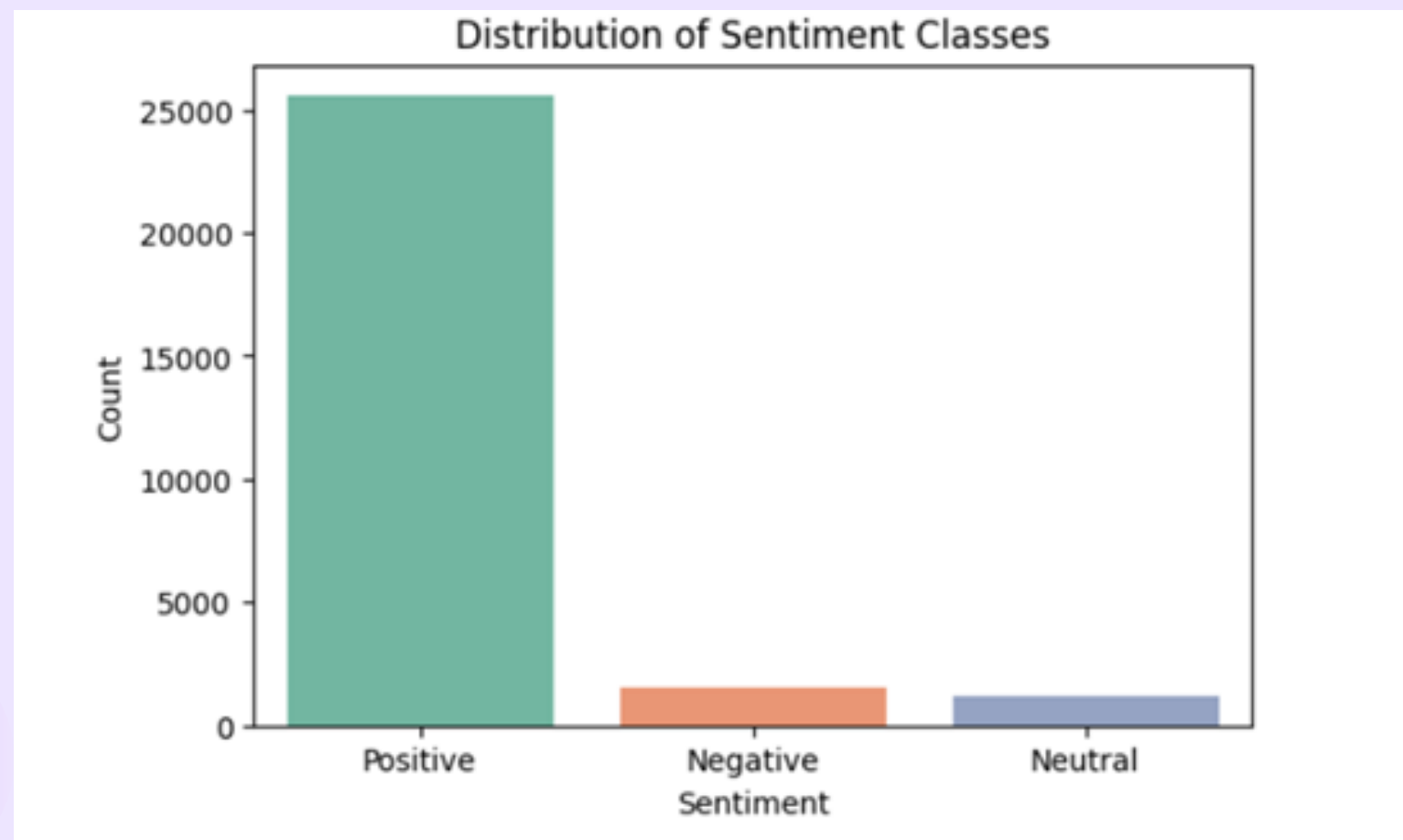
Reviews were classified into three categories using rating classification:

- Negative: Rating 1 or 2.
- Neutral: Rating 3.
- Positive: Rating 4 or 5.

Data preprocessing

● 3- Data Splitting

The data was split into a training set (80%) and a test set (20%) using `train_test_split` to ensure balanced data distribution.



● 4- Handling Imbalanced Data

Due to the class imbalance in the dataset, several techniques were used to address this challenge:

1. **Class Weights:** Adjusted the loss function to give more importance to underrepresented classes.
2. **Oversampling:** Duplicated samples from minority classes to balance the dataset.
3. **Undersampling:** Reduced samples from overrepresented classes to prevent bias.
4. **Data Augmentation:** Used NLP techniques to generate diverse variations of existing samples and enrich the dataset.



02

Classification of Reviews

Classification of Reviews



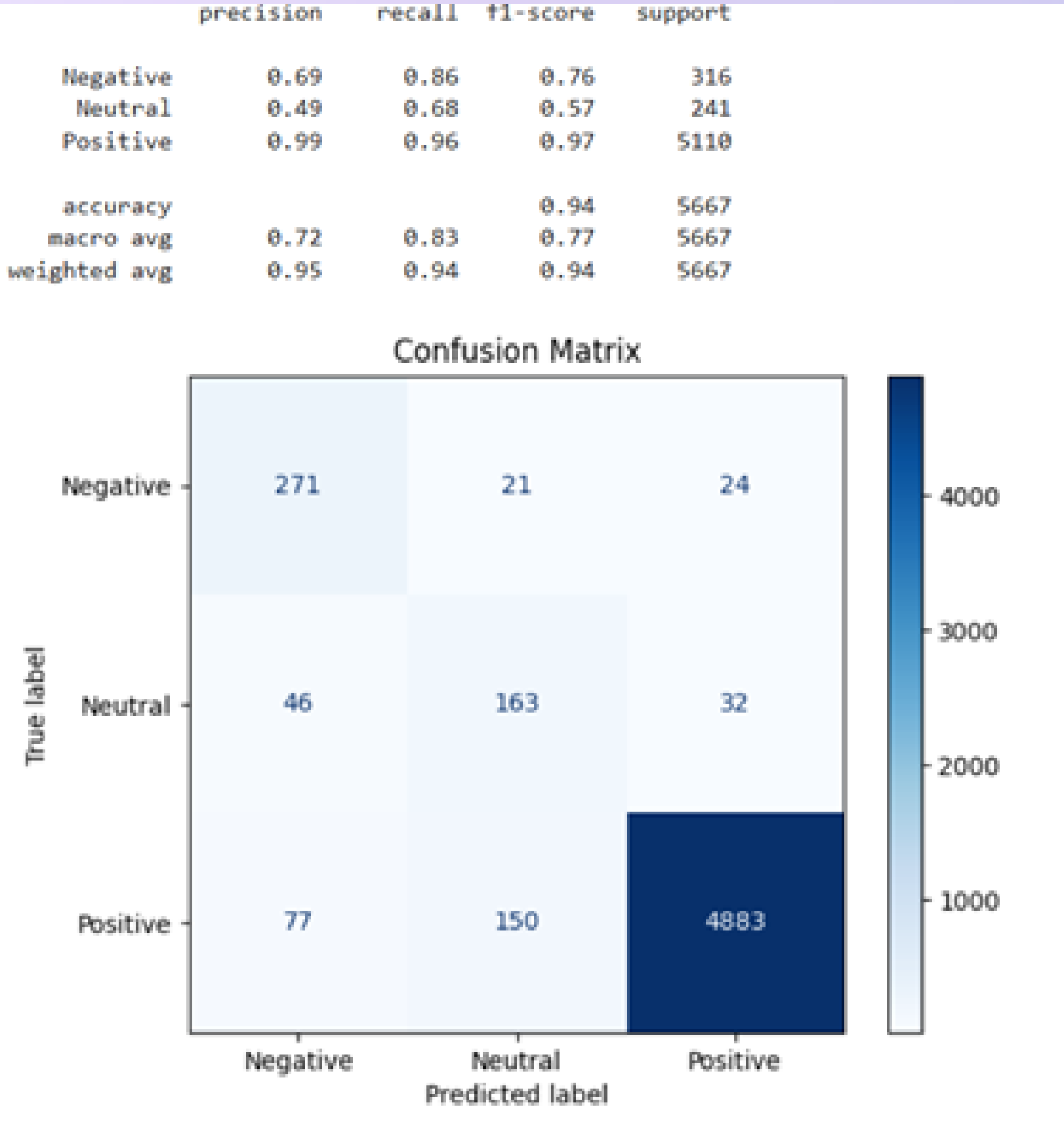
● Model Training

- The **DistilBERT model** from Hugging Face was fine-tuned to perform sentiment classification on the reviews.
- **TrainingArguments and Trainer** from the Hugging Face library were used to train the model and evaluate its performance.

● Evaluation

- The model's performance was evaluated using `classification_report`, which provides **accuracy, recall, and F1-score for each sentiment class**.
- A **Confusion Matrix** was also used to examine the distribution of predictions across the different classes.

● Model Performance





03

Clustering of Reviews

Clustering of reviews

● 1- Data Preparation

Generate the
DistilBERT
embeddings for
the product names

● 2- Clustering

- Using **KMeans** with a specified number of clusters, we found that 5 clusters was optimal based on the data.
- The clustering performance was evaluated using the **Silhouette Score**, which resulted in 0.17.

Clustering of reviews

● 3- Naming the Clusters

Based on the most frequent keywords and product types found within each group. These names were:

Cluster 0: "**Pet & Home Essentials**"

Cluster 1: "**E-Readers & Kids Tablets**"

Cluster 2: "**Smart Home Devices**"

Cluster 3: "**Fire Tablets**"

Cluster 4: "**Accessories**"

Clustering of reviews

● 4- Evaluation Metrics

The cosine similarity is used to evaluate the clustering quality.

- **Cluster 0: "Pet & Home Essentials":**
Cosine Similarity = **0.68**
- **Cluster 1: "E-Readers & Kids Tablets":**
Cosine Similarity = **0.71**
- **Cluster 2: "Smart Home Devices & Accessories"**
Cosine Similarity = **0.68**
- **Cluster 3: "Fire Tablets"**
Cosine Similarity = **0.51**
- **Cluster 4: "Accessories"**
Cosine Similarity = **0.44**



04

Summarization of Reviews

Summarization of Reviews

After clustering products, top-rated items were selected based on positive reviews. Common complaints were extracted from negative reviews using **TF-IDF**, and the worst product per category was identified.

This data was sent to **ChatGPT** to generate clear and engaging blog-style summaries that highlight top products, address user concerns, and explain underperformance.



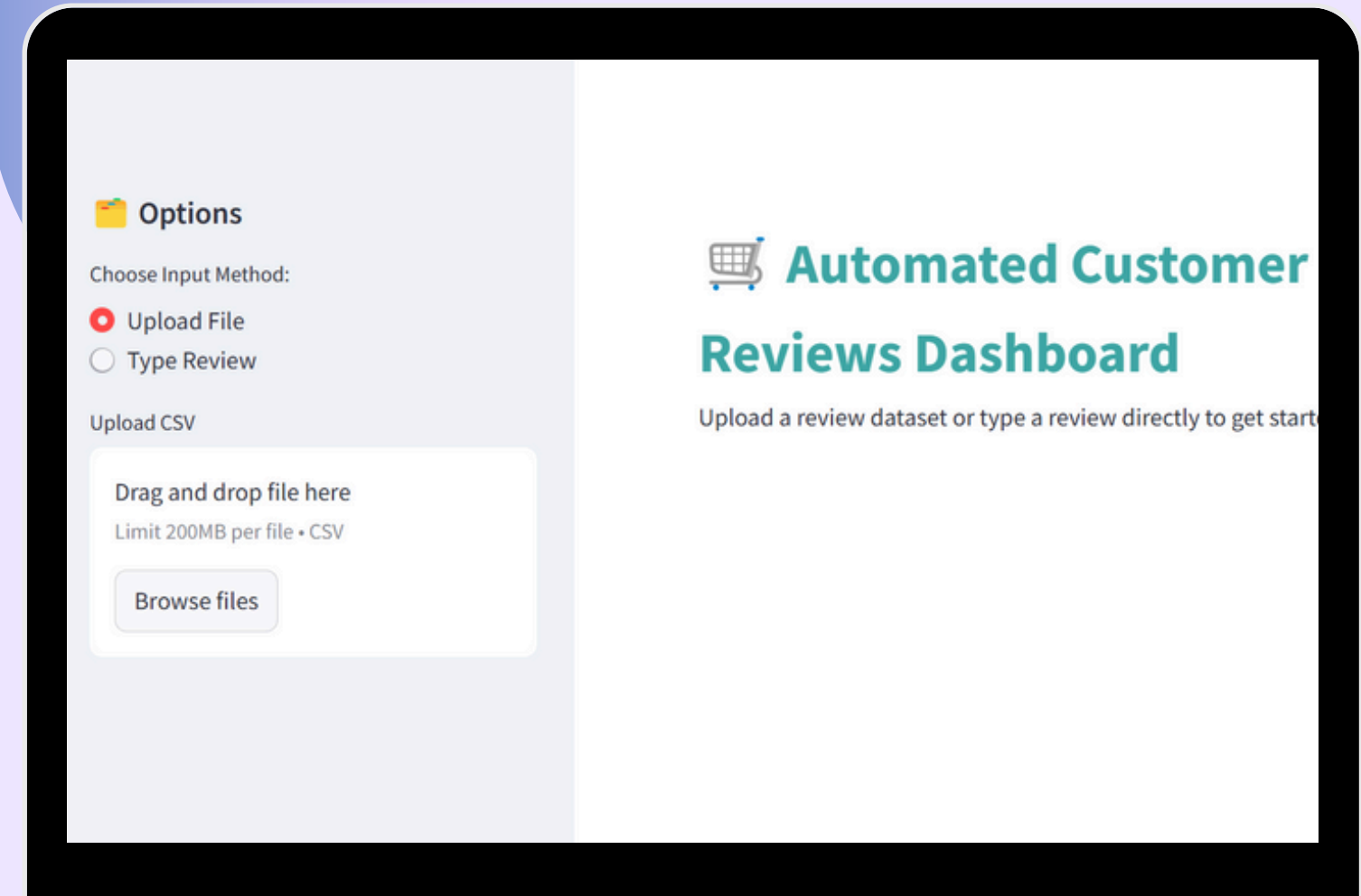
05 Deployment

Deployment

Use **Streamlit** for deploy the Automated Customer Reviews app



Streamlit





Thank You.

Any Questions?